app
Handover
RLF
Call Drops
RACH
Throughput
Beamforming
RCA Report
RF Coverage Map
VoNR
ANR
Config Audit
gNB Syslog
Alarm Correlation
Assurance Hub
UE Protocol
Upload
RF Coverage
View less

Navigation

Pages: Handover · RLF · Call Drops · RACH · Throughput · Beamforming · VoNR · ANR · Config Audit · gNB Syslog · Alarm · Assurance Hub · Upload · RCA Report

Focus cell

XYZ401

📡 XYZ Telecom Network Intelligence

AI-Powered Telecom Engineering & NPI Validation Copilot · Demo cells XYZ401–XYZ410 · Synthetic telecom datasets

Cells monitored	Cluster avg health	Worst cell	Cells grade D/C
10	53.1/100	XYZ407	10

🤖 AI-Powered RCA Workflow

Multi-Agent Root Cause Analysis

TNIC coordinates **12 specialist agents** and a **Master RCA orchestrator** to turn KPIs, drive-test data, syslog, CM, and alarms into ranked root causes, correlated impacts, and operator-ready action plans — with full agent traceability.

End-to-end workflow

1. **Query & KPIs** — PM · HO · RLF · assurance →
2. **Classifier** — 28 RCA types →
3. **Agent fan-out** — 12 specialists →
4. **Enrichment** — Coverage · UE · FM
5. **Correlation** — Cross-domain →
6. **Ranking** — Confidence score →
7. **Report** — Narrative + checklist

Specialist AI agents

🔀 **Handover**
Mobility
HO prep/exec failures, too-early/late, ping-pong

📡 **RLF**
Radio
Out-of-sync, T310 expiry, coverage-driven RLF

📵 **Call Drop**
Session
Drop classification — RF vs core vs mobility

📶 **RACH**
Access
PRACH detection, Msg3, contention failures

⚡ **Throughput**
Capacity
PRB, MCS, BLER, scheduler bottlenecks

📡 **Beamforming**
Massive MIMO
SSB utilization, beam gaps, switch stress

📞 **VoNR**
Voice
5QI-1 bearer, MOS, IMS/SMF session path

🔗 **ANR**
Neighbors
PCI conflict, missing NCL, ANR add/remove

⚙️ **Config Audit**
CM
Golden vs live drift — A3, TTT, PRACH

📋

🚨

📱

gNB Syslog

RAN logs

DU/CU, NGAP, XnAP, F1 transport faults

Alarm Correlation

FM

Critical alarm chains, transport vs RF

UE Protocol

UE trace

PHY→NAS failure stage localization

🗺️

RF Coverage

Geospatial

RSRP/SINR holes, beam congestion, drive test

How evidence is weighted

Scoring model

BASE SCORE **Rule confidence**

Each fired rule carries a calibrated confidence (0–1) from threshold severity and KPI deviation.

BOOST **Primary domain +10%**

Findings in the classified issue domain (e.g. HO for handover queries) rank higher.

BOOST **Classifier +15%**

Drop/HO classifiers add evidence when KPI patterns match known failure modes.

SUPPORTING **UE trace −12%**

UE protocol findings support the narrative unless the query is UE-specific.

ENRICHMENT **Assurance datasets**

Syslog, CM, ANR, VoNR, and FM alarms inject cross-domain evidence blocks.

CAUSAL CHAIN **Coverage correlation**

RF holes propagate to HO, RLF, RACH, throughput, and VoNR findings.

How agents correlate findings

Coverage → mobility chain: RF holes at cell edge drive HO prep failures, RLF, RACH access issues, throughput collapse, and VoNR bearer instability.

Workflow templates: Call-drop, HO-failure, RACH, throughput, VoNR, and cell-outage workflows inject cross-domain checks (ANR neighbors, CM drift, transport).

Assurance fusion: gNB syslog, config audit, FM alarms, and UE protocol traces are merged into a single evidence graph for the Master RCA narrative.

How Master RCA determines root cause

1. Classify — Parse query + KPIs; detect issue type and RCA catalog entry.
2. Orchestrate — Fan out to domain-specific agent chain (e.g. HO → RLF → ANR → syslog → alarm → RF coverage).
3. Enrich — `enrich_master_rca()` adds coverage correlation, workflow blocks, assurance evidence, and UE failure localization.
4. Rank — Sort by adjusted confidence: primary-domain +10%, classifier +15%, UE supporting −12%.
5. Deliver — Top-5 probable causes, deduplicated actions, validation checklist, health score, and optional LLM narrative report.

Example RCA journey — XYZ401

1. **Operator query**
   "Handover failure cell XYZ401" — KPIs merged from PM, HO, RLF, and assurance datasets.
2. **Issue classifier**
   Detects primary domain `handover`; selects agent chain from 28-type RCA catalog.
3. **Multi-agent fan-out**
   HO, RLF, ANR, syslog, alarm, and RF coverage agents run in parallel.
4. **Evidence enrichment**
   Coverage correlation links weak RSRP → HO prep failure; assurance adds CM drift.
5. **Master RCA ranking**
   Top cause: Coverage Deficiency (52/100) + Beam Congestion — confidence 94%.
6. **Action plan**
   Retilt sectors, fill holes, rebalance SSB beams 3–4; validate HO/RACH KPIs post-fix.

Current demo scope

- 10 demo cells (XYZ401–XYZ410) with synthetic PM, HO, RLF, RACH, throughput, and assurance data

- 12 specialist agents + Master RCA orchestrator with 28-type RCA catalog
- Handover enrichment layer (33-column RCA-ready dataset, 18 HO rules)
- RF Coverage geospatial agent with Plotly heatmaps and coverage-hole detection
- Upload & classify pipeline — CSV, syslog, UE trace → ingest → dynamic RCA
- Streamlit multipage dashboard + FastAPI (`/api/v1/analyze/rca`)

🌍 Real-World Deployment Architecture

How the current proof-of-concept evolves into a production telecom solution — from simulated datasets to predictive A operations.

Current State

Simulated telecom datasets

↓

CSV uploads

↓

Multi-agent RCA

↓

Recommendations

↓

Phase 1 — Field Log Ingestion

- TEMS logs
- Nemo logs
- QXDM traces
- UE log exports

↓

Phase 2 — OSS / Performance Integration

- PM counters
- Alarm feeds
- Daily KPI exports
- Drive test results

↓

Phase 3 — Live Operations Integration

- gNB syslog streaming
- Performance monitoring
- Alarm streaming
- Live KPI feeds

↓

Phase 4 — Predictive AI Operations

- Early warning detection

- Capacity risk prediction
- Coverage risk prediction
- Auto RCA
- Automated recommendations

Architecture note: The current platform validates AI-agent orchestration and RCA logic using representative telecom datasets. The architecture is intentionally **data-source agnostic** — future OSS and network integrations require minimal redesign.

🎯 How Confidence Scores Are Calculated

Confidence is **not a random AI value**. It is derived from evidence quality, supporting agents, data consistency, KPI severity, and historical rule matching.

① Evidence quality | ② Supporting agents | ③ Data consistency | ④ KPI severity | ⑤ Rule matching

Scoring model

Base confidence
Rule engine threshold score

+15%
Query classification

+10%
Primary domain match

+20%
Cross-domain evidence

+15%
gNB syslog validation

+15%
UE trace validation

+10%
Transport validation

+10%
RF correlation

+5%
Configuration validation

Example — Handover failure

Agent	Finding	Confidence
Handover Agent	Prep failure	78%
gNB Syslog	HO_REQUEST_ACK_TIMEOUT	99%
Transport Agent	**Xn latency high**	**97%**
RF Coverage Agent	Coverage healthy	95%

Primary RCA: Xn transport latency

Final confidence: **98%** — syslog + transport agree; RF contradicts coverage-hole hypothesis.

Confidence ranges

60–75%	**Low Confidence**	Single domain evidence
75–90%	**Medium Confidence**	Multiple supporting indicators
90–98%	**High Confidence**	Strong multi-agent correlation
98–100%	**Very High Confidence**	Cross-domain validation

⚖️ How the System Handles Conflicting Evidence

Telecom datasets often contain **contradictory indicators**. The Master RCA agent does not immediately trust a single agent — it reconciles evidence across domains before selecting a root cause.

1. Collect evidence from all specialist agents
2. Identify contradictions across domains
3. Rank evidence sources by quality and confidence
4. Evaluate network-side vs RF vs UE indicators
5. Select most probable cause with supporting consensus
6. Assign final confidence score

Example 1 — HO failure with conflicting RF evidence

Source	Finding	Confidence
Handover Agent	Coverage hole	82%
Transport Agent	**Xn latency**	**97%**
gNB Syslog	**HO_REQUEST_ACK_TIMEOUT**	**99%**
RF Coverage Agent	Coverage healthy	95%

Master RCA decision
Primary root cause: **Xn transport latency**

Two high-confidence network-side sources agree (transport + syslog). Coverage evidence is contradicted by RF measurements.
Final confidence: 98%

Example 2 — VoNR drop with layered causes

Source	Finding
VoNR Agent	**SIP timeout**
Transport Agent	Packet loss
UE Agent	QoS flow release

Master RCA output
Primary root cause: **IMS session timeout**
Secondary root cause: **Transport packet loss**
Final confidence: 94%

Executive Summary

Why Multi-Agent RCA Works

✓ Reduces false positives

✓ Validates findings across multiple domains

✓ Combines RF, mobility, protocol, core, and transport evidence

✓ Provides explainable recommendations

✓ Delivers higher confidence than single-domain analytics

Traditional KPI dashboard	TNIC multi-agent RCA
≈ One-domain visibility	**≈ End-to-end network intelligence**

Architecture reference: docs/RCA_AGENT_END_TO_END_HANDOVER.md · Live RCA: sidebar **RCA Report** or **Assurance Hub**

AI-Powered Telecom Engineering & NPI Validation Copilot

TNIC evolves beyond RCA dashboards into an end-to-end platform for dataset ingestion, multi-agent analysis, change impact assessment, and deployment readiness validation.

🧪 Current Synthetic Dataset Framework

Current RCA capabilities are validated using **representative telecom datasets** that mirror field log structure, OSS export and protocol trace formats.

📡

Mobility Dataset

`handover_events.csv`

CONTAINS

- HO Attempts
- HO Successes
- HO Failures
- Prep Failures
- Exec Failures
- Xn Failures

✅ Handover Agent · ✅ Transport Agent · ✅ RLF Agent

📻

RLF Dataset

`rlf_events.csv`

CONTAINS

- T310 Expiry
- N310 Events
- RLF Count
- Re-establishment Failures

✅ RLF Agent · ✅ RF Agent · ✅ UE Agent

📱

UE Protocol Dataset

`ue_protocol_trace.csv`

CONTAINS

- RACH
- RRC
- Registration
- Paging
- PDU Sessions
- VoNR Events

✅ UE Protocol Agent · ✅ VoNR Agent

📜

gNB Syslog Dataset

`gnb_syslog.csv`

CONTAINS

- HO_PREP_FAIL
- DRB_SETUP_FAIL
- XN_TIMEOUT
- UPF_UNREACHABLE

✅ gNB Syslog Agent · ✅ Transport Agent

☎️

VoNR Dataset

`vonr_sessions.csv`

CONTAINS

📊

RF Dataset

`enhanced_geospatial_rf_dataset.csv`

CONTAINS

- IMS Registration
- SIP Transactions
- VoNR Drops
- Call Setup Metrics

✅ VoNR Agent · ✅ UE Agent

- RSRP
- RSRQ
- SINR
- CQI
- Geo Coordinates

✅ RF Coverage Agent · ✅ Beamforming Agent

🚀 Future Data Source Expansion

Future-ready architecture — same ingestion pipeline and agent orchestration, expanded data adapters.

📱
UE Sources

FUTURE UPLOADS

- QXDM Logs
- TEMS Logs
- Nemo Logs
- Android Bugreports
- Protocol Traces
- PCAP Files

Future agent: UE Trace Agent

📡
gNB Sources

FUTURE UPLOADS

- CU Logs
- DU Logs
- gNB Syslogs
- NGAP Logs
- F1 Logs
- Xn Logs
- Crash Dumps

Future agent: gNB Deep Trace Agent

⚙️
Configuration Sources

FUTURE UPLOADS

- CM Snapshots
- XML Exports
- Neighbor Dumps
- PCI Plans
- OSS Config Exports

Future agent: Configuration Drift Agent

📊
OSS Sources

FUTURE UPLOADS

- PM Counters
- KPI Reports
- Alarm Exports
- Performance Reports

Future agent: Performance Analytics Agent

☎️
Core Sources

FUTURE UPLOADS

- AMF Logs
- SMF Logs
- UPF Logs
- IMS Logs
- SIP Traces

Future agent: Core Network RCA Agent

📥 Automated Data Ingestion Pipeline

Dataset Arrival ↓ Classifier Engine ↓ Metadata Extraction ↓ Telecom Domain Mapping ↓ RCA Trigger Engine ↓ AI Specialist Agents ↓ Master RCA Agent ↓ Recommendations

The same workflow supports:
✅ Synthetic datasets · ✅ Lab data · ✅ Field logs · ✅ OSS exports · ✅ Future real-time network data

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📂 Dataset Upload & Agent Simulation

Upload a telecom file or select a synthetic dataset to run the full **classify → trigger → agents → evidence → Master RCA** workflow — no live OSS integration required.

Upload dataset

Drag and drop file here
Limit 200MB per file • CSV, XLSX, XLS, TXT, LOG, JSON, XML, ZIP

Browse files

Or select synthetic sample dataset

Mobility Dataset

▶ Run Agent Simulation

Dataset → Classification → Trigger Engine → AI Agents → Evidence → Master RCA → Recommendations

Dataset classification

Source file	Detected type	Class confidence	Events parsed
handov…	RF_MEA…	75%	1000

Detected telecom domain

Mobility / Handover

Triggered agents

✅ Handover Agent ✅ RLF Agent ✅ PM Agent ✅ Latency Agent ✅ Transport Agent ✅ ANR Agent ✅ gNB Syslog Agent ✅ Alarm Agent ✅ UE Protocol Agent ✅ VoNR Agent

Agent activity panel

Triggered agents	Confidence	Evidence count
11	91%	15

Evidence repository

Agent	Finding	Confidenc
Handover Agent	[ANR correlated] ANR missing neighbor / PCI conflict events	91%
Handover Agent	[Coverage correlated] Coverage hole — HO prep/exec fails at cell edge	86%
gNB Syslog Agent	Syslog evidence: 350 events — top codes: HO_PREP_FAIL(76), MSG1_FAIL(63), SIP_TII	95%
ANR Agent	ANR evidence: 178 events — PCI conflict=59, missing neighbor=48, add fail=71	95%
Master Rca	Assurance evidence summary: 21 blocks, mean confidence 86%, types=['alarm', 'anr',	95%
Alarm Agent	Alarm correlation: 162 active alarms (52 CRITICAL) — Transport Packet Loss(70), Bean	94%
RF Coverage Agent	Coverage Deficiency: Retilt sector A/B, fill edge coverage holes, rebalance SSB beams	94%
Handover Agent	[Syslog correlated] Syslog HO_PREP_FAIL / XN_TIMEOUT (122 events)	83%
VoNR Agent	VoNR session evidence: 168 sessions, drop rate 45.24%, causes={'QOS_FLOW_FAIL': 3	92%
Handover Agent	High HO preparation failure — target cell not ready or Xn/NG timeout	82%

Master RCA output

Primary root cause: [ANR correlated] ANR missing neighbor / PCI conflict events
Secondary root cause: [Coverage correlated] Coverage hole — HO prep/exec fails at cell edge

91% confidence

Recommendations:

- Run handover RCA after ANR fix
- Resolve coverage on XYZ401 to clear HO Failure
- Check Xn/NG HO prep path and neighbor readiness
- Map RLF cluster geography and correlate with RSRP/SINR
- Audit PRACH configuration and UL interference
- Rebalance beam load and review scheduler weights

⚡ Agent Trigger Engine

Automatically triggers relevant specialist agents based on dataset classification and detected KPI anomalies.

Example 1

`Input: handover_events.csv`

Detected:

• HO Success Rate = 89%

• Prep Failure = 7%

Agents triggered: ✅ Handover Agent · ✅ Transport Agent · ✅ RLF Agent

Example 2

`Input: ue_protocol_trace.csv`

Detected:

• Registration Reject

• PDU Failure

Agents triggered: ✅ UE Protocol Agent · ✅ gNB Syslog Agent

Example 3

`Input: vonr_sessions.csv`

Detected:

• VoNR Drop Increase

Agents triggered: ✅ VoNR Agent · ✅ Transport Agent · ✅ UE Protocol Agent

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🔄 Multi-Dataset Correlation

Example inputs: handover_events.csv · gnb_syslog.csv · transport_metrics.csv · ue_protocol_trace.csv · neighbor_snapshot.xml

↓

📡 Handover Agent | 🌐 Transport Agent | 📜 Syslog Agent | 📱 UE Agent | ⚙️ Config Agent

↓

🧠 Master RCA

Different agents consume different datasets simultaneously. All findings are combined by the Master RCA agent.

🤖 Agent Activity Monitor

handover_events.csv	11	82	91%
Dataset	Triggered agents	Evidence generated	Confidence

Last analysis (XYZ401): [ANR correlated] ANR missing neighbor / PCI conflict events

✅ Handover Agent · ✅ Transport Agent · ✅ Syslog Agent · ✅ RLF Agent
✅ VoNR Agent · ⚪ Beamforming Agent

🚀 NPI Validation Center

Purpose: Validate telecom network changes before and after deployment — software upgrades, feature activations, hardware swaps, and customer acceptance testing.

✅ Software Upgrades · ✅ Feature Activations · ✅ Hardware Changes · ✅ Beamforming Enhancements
✅ Scheduler Changes · ✅ Mobility Parameter Updates · ✅ VoNR Feature Rollouts · ✅ Customer Acceptance Testing

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📊 Change Impact Analysis

Before Dataset + After Dataset → AI Comparison Engine → Agent Analysis → Regression Detection → Deployment Recommendation

Example — Software upgrade 24R1 → 24R3
✅ Throughput +14% ✅ VoNR setup success +2%
⚠ HO prep failure +5%
Risk: Medium · **Recommendation:** Proceed with limited rollout

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✅ Deployment Readiness Assessment

Outputs: **PASS** · **PASS WITH RISKS** · **FAIL**

PASS WITH RISKS

Release: 24R3
✅ Throughput improved ✅ VoNR improved
⚠ Mobility regression detected
Confidence: 95%

🌍 Real-World Deployment Roadmap

🧪 **Current PoC**
Synthetic datasets · CSV upload · multi-agent RCA · recommendations

↓

📱 **Field Logs**
TEMS · Nemo · QXDM · UE log exports

↓

🏢 **OSS Integration**
PM counters · alarm exports · syslog feeds

↓

🚀 **Automated AI Operations**
Streaming PM · live syslogs · predictive analytics · auto RCA

Executive message: The current PoC validates AI orchestration, RCA logic, confidence scoring, and agent collaboration using synthetic telecom datasets. **Future phases replace the data source, not the AI architecture.**

Key Value to NPI Engineers

Traditional workflow

- Manual KPI analysis
- Manual log analysis
- Manual before/after comparison
- Multiple disconnected tools

AI-powered workflow

Upload data → Automatic classification → Specialist agents → Evidence correlation → Master RCA → Deployment recommendation

- ✅ Faster Troubleshooting
- ✅ Regression Detection
- ✅ Change Impact Analysis
- ✅ Explainable RCA
- ✅ Software Validation
- ✅ Feature Validation
- ✅ Customer Acceptance Readiness

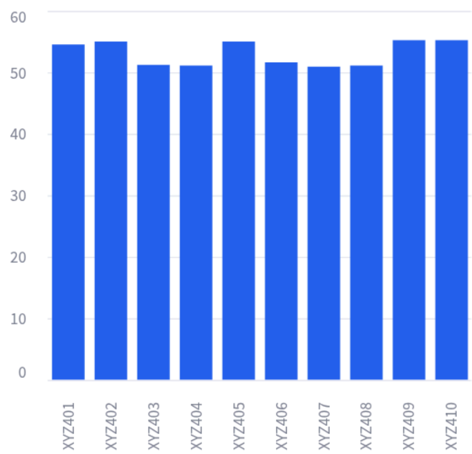
Final vision: TNIC evolves from a telecom RCA dashboard into an AI-assisted NPI Validation Copilot — analyzing synthetic datasets today and future UE logs, gNB logs, CM snapshots, PM counters, OSS exports, alarms, and real network telemetry tomorrow.

NPI Copilot · Upload & Analyze: sidebar **Upload** · Live RCA: **RCA Report** · All specialist agents: **Assurance Hub**

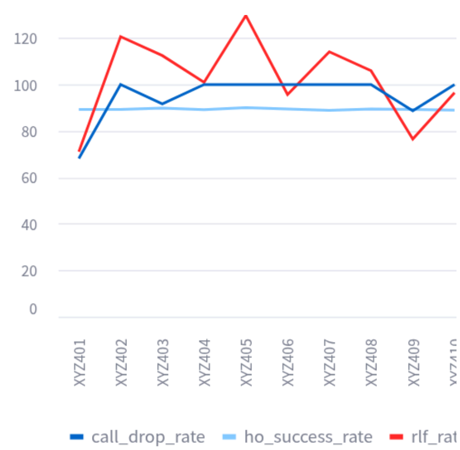
Network health overview

cell_id	health_score	grade	ho_success_rate	rlf_rate	call_drop_rate	rach_success_rate	throughput_ml
XYZ407	50.9	D	88.87	114.12	100	18.37	475
XYZ404	51.1	D	89.23	100.97	100	17.05	553
XYZ408	51.1	D	89.51	105.95	100	20	494
XYZ403	51.2	D	89.89	112.5	91.67	20.34	536
XYZ406	51.6	D	89.5	95.7	100	25.53	558
XYZ401	54.5	D	89.3	71.11	68.15	23.58	528
XYZ402	55	D	89.31	120.65	100	19	5
XYZ405	55	C	90.03	129.76	100	24.24	480
XYZ409	55.2	C	89.29	76.52	88.7	19	485
XYZ410	55.2	C	89.02	96.46	100	21.65	491

Health score by cell



Mobility & reliability



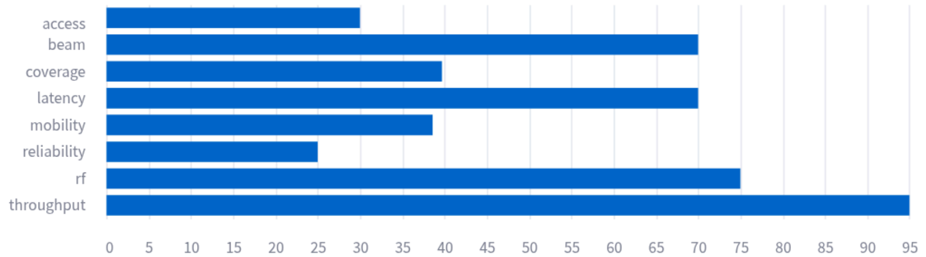
RF & throughput snapshot

cell_id	ss_rsrp	ss_sinr	throughput_mbps	rach_success_rate
XYZ401	-105.17	5.27	528.96	23.5
XYZ402	-103.58	5.28	515	1
XYZ403	-102.54	3.28	536.24	20.3
XYZ404	-103.47	4.68	553.61	17.0
XYZ405	-104.16	5.14	480.78	24.2
XYZ406	-102.36	4.93	558.91	25.5
XYZ407	-103.14	3.92	475.65	18.3
XYZ408	-102.39	3.1	494.56	2
XYZ409	-102.72	6.47	485.35	1
XYZ410	-102.41	5.2	491.38	21.6

Focus cell — XYZ401

Health score Grade Alerts

54.5/100 D 2



Mobility health critical — review HO KPIs

Drop rate elevated — run call drop RCA

Data: PM/HO/RLF/RACH + assurance datasets (syslog, CM, ANR, VoNR, alarms)